ICT305 Topics in IT

Robotic Process Automation
and Smart Cities

CIHE Australia

Robotic Process Automation

Agenda

- Definition of Robotic Process Automation (RPA)
- Motivation and importance of RPA
- Differences between Test Automation and RPA
- RPA implementation methodology
- Best practices of RPA implementation

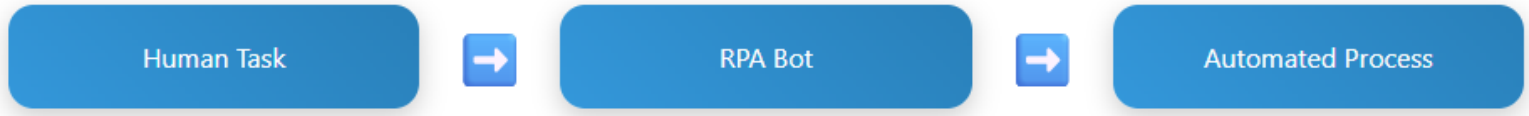
Definition of robotic process automation (RPA)

- **R**obotics **P**rocess **A**utomation (**RPA**) allows organizations to automate task just like a human being was doing them across application and systems.

- **RPA** can be used to automate workflow, infrastructure, back-office process which are labor intensive.

- **These software bots** can interact with an in-house application, website, user portal, etc.

- The main motivations to go for RPA are most often:
  - the shortage of IT resources, which leads business departments to build things on their own (hoping to get along without IT)
  - the refusal to invest in a proper modernization strategy of the enterprise IT.



Importance of RPA

How RPA Works

- **Technical Implementation:**
  - ✓ Screen scraping and data extraction
  - ✓ API integration and web services
  - ✓ Database connectivity and manipulation
  - ✓ Email and document processing
  - ✓ Exception handling and error management
  - ✓ Workflow orchestration and scheduling

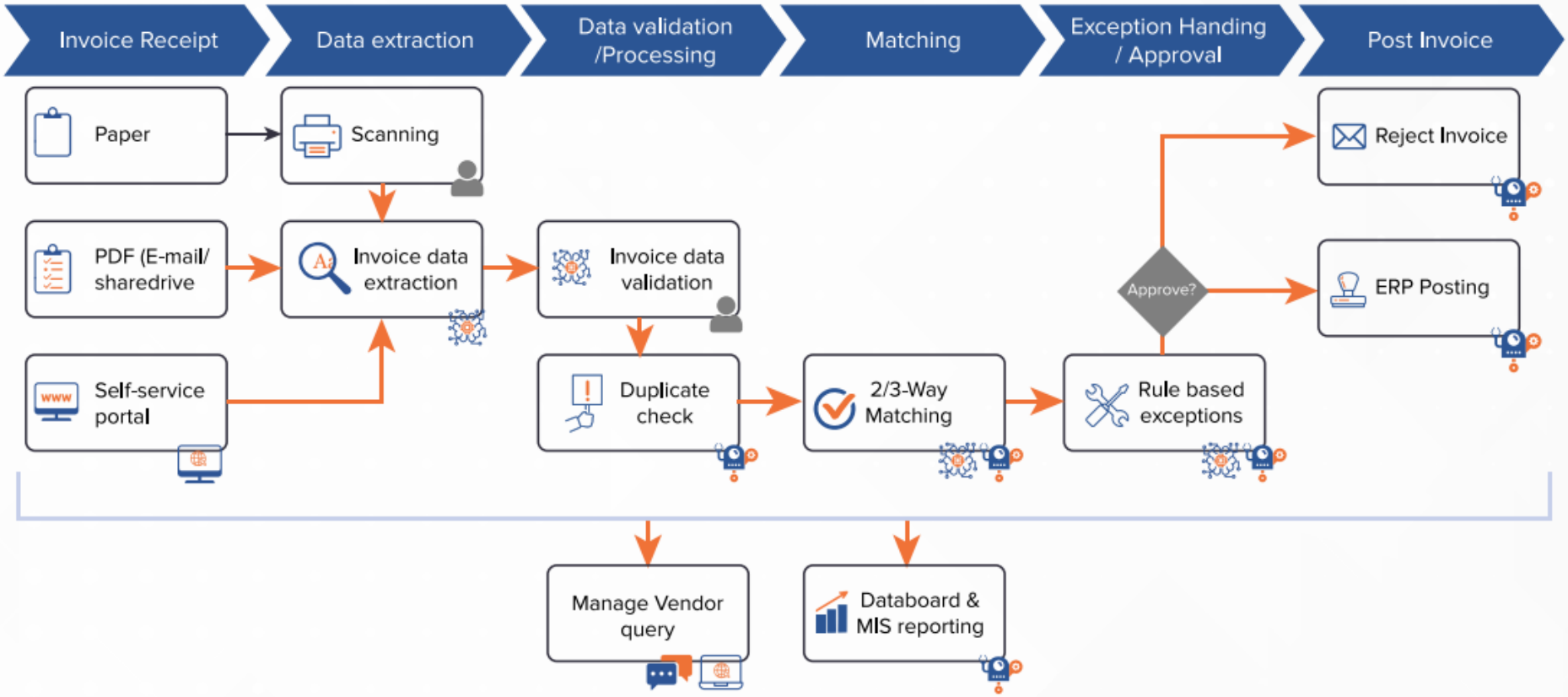
Example: Processing invoice data from emails → extracting information → updating ERP system → sending confirmation

RPA vs Traditional Automation

Aspect	RPA	Traditional Automation
Implementation	Works through UI layer	Requires system integration
Setup Time	Days to weeks	Months to years
System Changes	No changes needed	Extensive modifications
Cost	Lower upfront investment	Higher initial costs
Flexibility	Easy to modify	Complex to change
Scalability	Medium	High

RPA is ideal for quick wins and legacy system integration

Example of RPA for invoice processing



Enhance Invoice Processing Using RPA - Nividous Intelligent Automation Company

Example of RPA

Description	Can be Automated via RPA?
Open invoice email from the supplier and print it for records	Yes
Barcode Scanning	Manual
Create work item in a legacy software system	Yes
Enter PO to retrieve Invoices	Yes
Check supplier name is correct or not?	Yes
Key Invoice, Data and Amount	Yes
Check if Amount is matches or not?	Yes
If amount match is yes the Matched Invoice, Calculate Tax	Yes
Complete Invoice Processing	Yes
Work Item Closed	Yes

Differences between Test Automation and RPA

Parameter	Test Automation	RPA
Goal	Reduce Test execution time through automation	Reduce headcount through automation
Task	Automate repetitive Test Cases	Automate repetitive Business processes
Coding	Coding knowledge required to create Test Scripts	Wizard-driven, and coding knowledge not required
Tech Approach	Supports limited software environment. Example: Selenium can support only web applications.	Supports a wide array of software environments
Example	Test cases are automated	Data entry, forms, loan processing, is automated
Application	Test Automation can be run on QA, Production, Performance, UAT environments	RPA is usually run only on production environments
Implementation	It can automate a product.	It can automate a product as well as a service.
Users	Limited to technical users.	Can be used across the board by all stakeholders.

RPA implementation methodology

- Planning/Analysis
- Development/Bot Development
- Testing
- Support & Maintenance

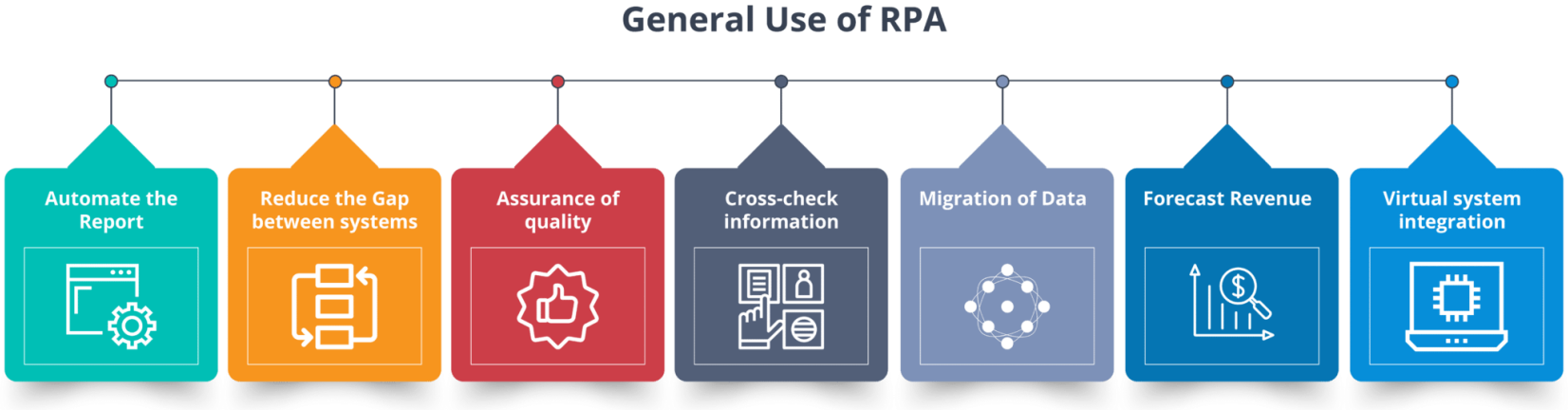
Best practices of RPA implementation

- One should consider **business impact** before opting for RPA process,
- Define and focus on the desired **ROI,**
- Focus on **targeting larger groups** and automating large, impactful processes,
- Combine attended and unattended RPA,
- **Poor design**, change management can wreak **havoc,**
- Don't forget the impact on people.

Best practices of RPA

- Create Process Design Documents
- Use Reusable Component
- Developed flow should be readable and not time-consuming
- Build Error Handling
- Break Process
- *Typical example*:
  - With RPA, online ticket bookings can be automated by getting booking information and upload it to the database and confirmed bookings to the user.

General use of RPA



Application of RPA

Healthcare

- Registration of Patients
- Maintaining Records of Patients

HR

- Joining Formalities of new Employees
- Process of Payroll

Telecom

- Service Order Management
- Quality Reporting

Insurance

- Processing & Clearance of Claims
- Providing Premium Information

Retail Industries

- Maintaining Bills of Materials
- Calculation of Sales

Travel & Logistic

- Booking of Tickets
- Maintaining Passenger Details

Robotic Process Automation tools

Selection of RPA Tool should be based on following 4 parameters:

- **Data**: Easy of reading and writing business data into multiple systems
- **Type of Tasks mainly performed**: Ease of configuring rules-based or knowledge-based processes.
- **Interoperability**: Tools should work across multiple applications
- **AI**: Built-in AI support to mimic human users
- **Popular Robotic Automation Tools**: Blue prism, Automation AnyWhere, UiPath etc..

Popular RPA Tools & Platforms

Ui

UiPath

Market leader with comprehensive automation platform

AA

Automation Anywhere

Cloud-native platform with AI capabilities

BP

Blue Prism

Enterprise-focused with strong security

MS

Microsoft Power Automate

Integrated with Office 365 ecosystem

WA

WorkFusion

AI-powered intelligent automation

PG

Pega

Process automation with case management

Benefits of RPA

Cost Reduction
30-50% reduction in operational costs

Speed
Faster processing times

Accuracy
99.9% accuracy in data processing

24/7 Operation
Continuous processing without breaks

Scalability
Easy to scale up or down based on demand

Employee Satisfaction
Eliminates repetitive tasks, focuses on value-added work

Disadvantages of RPA

- The bot in RPA is limited to the speed of the application.
- Even small changes made in the automation application will need the robots to be reconfigured.

Future of RPA

- **Intelligent Process Automation (IPA):**

RPA + AI/ML + Analytics = IPA

🧠 **AI Integration**

- ✓ Natural Language Processing
- ✓ Computer Vision
- ✓ Machine Learning decisions
- ✓ Predictive analytics

🔮 **Emerging Trends**

- ✓ Hyperautomation
- ✓ Low-code/no-code platforms
- ✓ Cloud-native solutions
- ✓ Process mining integration

Smart Cities

Agenda

1. Motivation
2. Definition
3. Technologies to support smart cities
4. Significance of smart cities
5. Issues of smart cities
6. Future of smart cities

Motivation

- 21st Century has brought with it a new global trend of “sustainable urban development” and this concept adds new dimensions to urbanization which require a quick need to upgrade existing cities.
- A “city service” can be anything a city offers to its citizens or businesses, including lighting, traffic management, public parking, electricity, etc.

Definition

- **A Smart City** is the integration of technology into a strategic approach to sustainability.
- **A Smart City** is an urban area that has become more efficient and/or more environmentally friendly and/or more socially inclusive through the use of digital technologies.
- **The goal of a Smart City** is to improve its attractiveness to citizens and/or businesses by enhancing and/or adding city services.
- Due to the absence of a universally accepted Smart City definition, quite diverse concepts have been filed under the term “Smart City” with a relatively high degree of freedom.

Smart city projects

Supporting technologies for smart cities

- Personalisation of services
- Data sharing
- IoT
- Internet of People
- Electrification
- Digital Twins
- Artificial intelligence (AI)
- Renewable energy sources, smart sustainability and waste solutions
- Quantum computing
- RPA

Personalisation of services

- Data is transformative as a way to understand, optimise and personalise services and infrastructure.
- Smart services will create a greater willingness of citizens to share their data with government and business in return for highly personalised services that improve their experience, help drive operational productivity and support the creation of new, high-value services.

Data sharing

- Governments have vast amounts of accumulated data (e.g demographic information about their citizens, records detailing public service usage).
- So too do private sector partners such as developers and transport operators.
- Mutual benefit and good governance drive greater data sharing.
- When shared and combined, these public and private data streams are essential to smart city design, as many of the foundational elements of smart precincts are achieved by joining up data to get valuable insights.*

IoT

- Serving as the backbone hardware for smart precinct developments, **emerging technologies such as IoT** are driving many smart precinct initiatives, while also evolving in the way they are used.
- IoT creates unique, user-centric experiences while also improving efficiency and sustainability in communities.
- Because IoT overlaps physical sectors such as food, water, energy and transport, data interoperability is key to support data flow and data federation.

Internet of People

- **Wearable fitness devices** promote positive health outcomes for citizens of a precinct.
- When paired with a smartphone or website to track performance, wearables have the added bonus of creating an 'Internet of People'.
- The popularity of wearables means an increased density of sensors and therefore more potential data sources.
- They effectively increase ways to connect people to places.

Electrification

- Electrification will enhance local mobility and reduce emissions within precincts.
- Long-term planning should factor in electric vehicles (cars, bicycles, scooters, etc.), plus the need for vehicle charging inbuilt to street or building parking.

Digital Twins

- **Digital twins** provide real-time updates and come with a variety of functions and uses such as building information models (BIM) to support planning and construction, and to plan movement within places.
- Smart precincts should use digital twins that extend beyond just ‘straightforward’ data lakes (or similar) to incorporate strategic planning tools - ‘what-if’ engines that allow you to model different scenarios and give you valuable insights to help you make key decisions.
- For example, if a new station is being built in a precinct, what other investment into surrounding infrastructure, services and amenities needs to be made to ensure that the place that is being built meets the current and future wants and needs of the different citizen demographics in the precinct?

Artificial intelligence (AI)

- AI will drive greater personalisation of services in smart precincts, optimising systems, augmenting decision-making and driving even greater use of data.
- Importantly, this growing capability is subject to ongoing standardisation efforts and requires community engagement around the **ethical and responsible uses of AI**.

Renewable energy sources, smart sustainability and waste solutions

- Renewable energy sources will become increasingly important in Australia's smart cities.
- So too will sustainability solutions such as vacuum waste systems, which could reduce emissions by 90 percent, and waste collection costs by 30 percent.
- In Western Sydney, for instance, this could save residents AU$62.8 million per year, and deliver indirect environmental and health benefits.

Quantum computing

- Quantum computing has the potential — in the next ten to twenty years — to bring the next great productivity uplift in scale and speed of computing.
- For smart precincts, this means the possibility of delivering more cost- and energy-efficient solutions to problems such as large scale and granular machine learning and optimisation, currently ‘impossible’ for today’s classical computers.

RPA in Smart Cities

🏛️ Government Services

- ✓ Permit processing
- ✓ License renewals
- ✓ Tax administration
- ✓ Citizen complaint handling

🚦 Traffic Management

- ✓ Traffic violation processing
- ✓ Parking citation management
- ✓ Infrastructure monitoring
- ✓ Emergency response coordination

💡 Utilities

- ✓ Smart meter data processing
- ✓ Billing automation
- ✓ Service request handling
- ✓ Outage management

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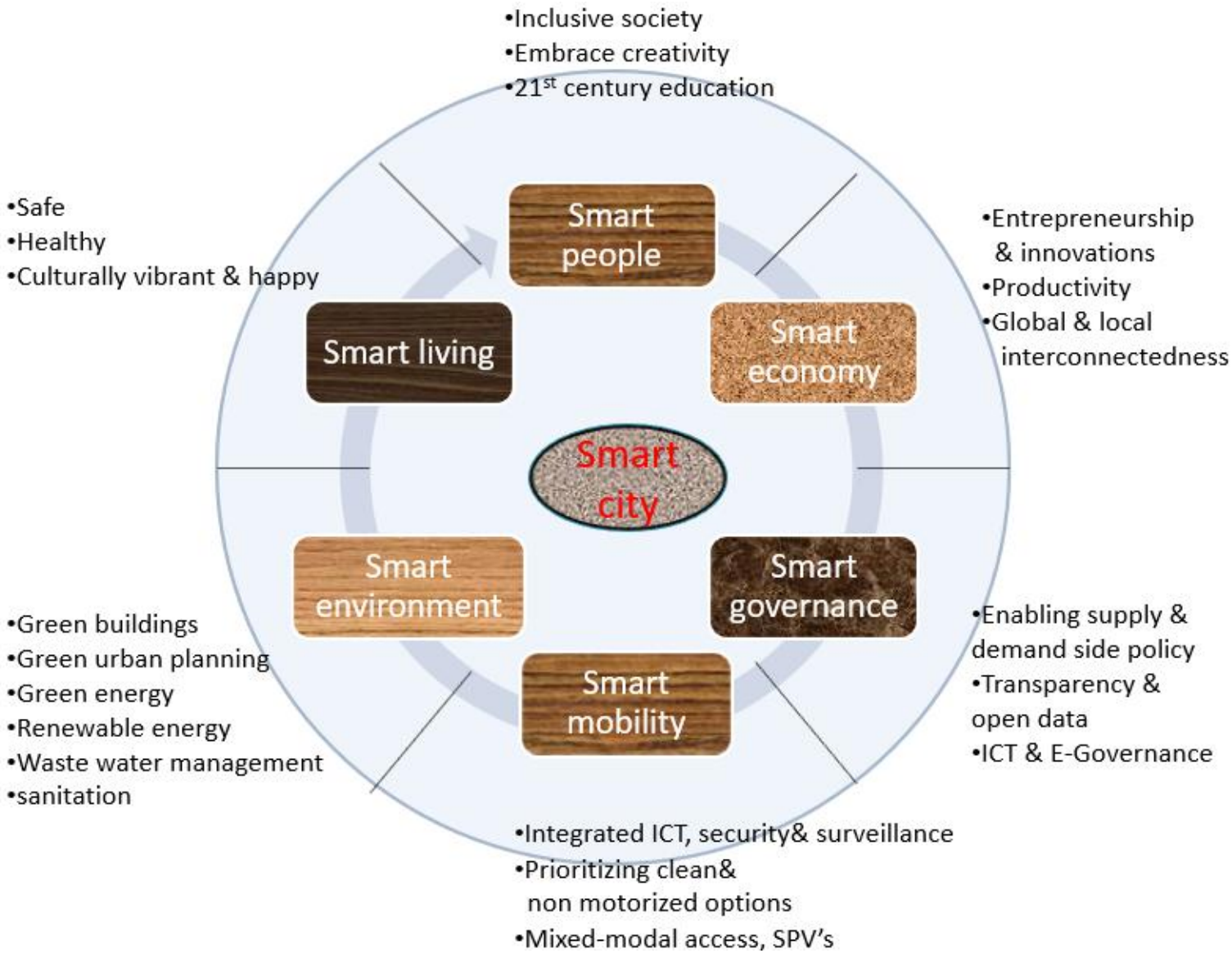
Significance of smart cities

- Economic development,
- Global city investment competitiveness,
- Improving lives and livelihoods of residents
- Proper facilities for entertainment, safety, security, & education are must,
- In terms of infrastructure, the smart cities should have 24*7 availability of water and electricity,
- Importance should be given to public transport,
- Energy efficiency techniques using 3R's concept
- Social infrastructure like parks, gardens, community halls for interaction in compactness,
- Employment opportunities ,
- Efficiency, sustainability, and livable.

Indicators of smart cities



Challenges for smart cities

- Foundation of a fundamental, standards-based IT infrastructure
- Smart city planning is a balancing act
- Smart cities need effective and efficient data process and analytics
- Smart cities must protect residents' data to assure privacy concerns
- Political differences can be a roadblock to smart city deployments

Future smart cities

Conclusion

- The emergency of the idea of the smart city ought to be seen in the backdrop of the need to contain corruption and meet the requirements of the adequate & efficient service delivery in urban areas using information and digital technologies.
- The success of the programmes will be judged by its power to transform the life of the people, and their ability to reduce growing inequality in our society.

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